One network for renders and video: vendor-neutral neural temporal super-resolution and low-latency frame generation trained on commodity cloud GPUs

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Abstract

Neural upscaling and frame generation now set image quality in many games, but the strongest commercial systems need one vendor's recent GPUs, and research systems need engine data that ordinary video lacks. We present NeuralSS, an open system that serves both cases with one model per task. Its temporal super-resolution network is a recurrent, re-parameterisable convolutional network with a built-in motion estimator that runs only when the engine supplies no motion vectors. Its frame-generation network fine-tunes the open RIFE 4.26 interpolation network, adds engine hints through zero-initialised channels, extrapolates with a late-latched camera prior and blends user-interface pixels towards the newest real frame. Both are trained in 10.13 and 10.07 hours on two NVIDIA T4 GPUs from public datasets, after copy detection that left no duplicate across splits, and both export to ONNX. On five test suites the 3x super-resolution student (16.1 GMAC per 1080p frame) beats bicubic upscaling on ordinary video by 1.19 to 2.19 dB PSNR-Y and an analytic temporal anti-aliasing upscaler by 2.79 to 4.91 dB with engine data, with lower temporal-change error. The largest frame-generation tier beats unchanged RIFE 4.26 on the Middlebury interpolation set (34.23 against 33.34 dB) and frame repetition in extrapolation by 1.86 to 5.57 dB, but the 12.7-GMAC tier is still 1.78 dB below RIFE on Middlebury, the cut detector false-alarms on multi-frame gaps, and eager PyTorch on a T4 needs 27.7 and 48.1 ms per 1080p frame, above real-time budgets. We report these limits with the gains.

Keywords: Temporal super-resolution, Frame generation, Video frame interpolation, Real-time rendering, Latency, Efficient neural networks

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